

Multiple Choice (10 marks)

1. What factors limit the number of trophic levels in a food chain?
 - (a) Constant energy levels across all trophic levels
 - (b) Increase in nutrient availability at each level
 - (c) Decrease in carbon dioxide concentration
 - (d) Increase in biomass at each successive level
 - (e) Great decrease (90%) in available energy at each level
2. An organism belonging to the nekton is which one of the following?
 - (a) Sponge
 - (b) Protist
 - (c) Clam
 - (d) Starfish
 - (e) Coral
3. Which of these is the smallest unit upon which natural selection directly acts?
 - (a) an individual's genome
 - (b) a population's genetic variation
 - (c) a species' genetic mutation
 - (d) an individual's genetic makeup
 - (e) a population's gene frequency
4. The first point of entry of water at the roots of a monocot plant is through the cytoplasm of cells of the
 - (a) parenchyma
 - (b) phloem
 - (c) endoderm
 - (d) pericycle
 - (e) cortex
5. What is an oncogene?
 - (a) Oncogenes are found only in viruses
 - (b) Oncogenes prevent cancer cells
 - (c) Oncogenes are a type of immune cell
 - (d) Oncogenes detoxify harmful substances in the cell
 - (e) Oncogenes slow down cell division

True / False (10 marks)

Answer True or False

6. True or False: Only inherited genetics and lifestyle factors such as diet and exercise can lead to the development of cancer.
7. True or False: Rotational cleavage is characterized by early cell divisions occurring at an oblique angle to the polar axis.
8. True or False: Sound communication in animals serves functions such as mate attraction, species recognition, defense, and territorial claim.
9. True or False: When drawing from a standard deck of cards, the probability of drawing the ace of diamonds is $1/52$.
10. True or False: The zona pellucida is responsible for enclosing terrestrial vertebrate embryos to protect them from desiccation and injury.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the adaptive features that animals possess to survive in hot arid environments, focusing on the role of the loops of Henle in water conservation.
12. Discuss the physiological mechanisms that ensure food and air are channeled into the correct pathways during swallowing, emphasizing the roles of the pharynx, larynx, and epiglottis.

End of Paper